## Symptom:
## P0731-GEAR RATIO ERROR IN 1ST

**When Monitored and Set Condition:**

**P0731-GEAR RATIO ERROR IN 1ST**

When Monitored: The Transmission Gear Ratio is monitored continuously while the Transmission is in gear.

Set Condition: If the ratio of the input RPM to the output RPM does not match the current Gear Ratio.

| POSSIBLE CAUSES |
|---|
| RELATED DTC'S PRESENT |
| INTERMITTENT GEAR RATIO ERRORS |
| INTERNAL TRANSMISSION |

| TEST | ACTION | APPLICABILITY |
|---|---|---|
| 1 | **NOTE: Low fluid level can be the cause of many Transmission problems. If the fluid level is low, locate and repair the leak, then check and adjust the fluid level per the Service Information.**<br>**NOTE: Always perform diagnostics with a fully charged battery to avoid false symptoms.**<br>With the DRBIII®, read the engine DTC's. Check and repair all engine DTC's prior to performing Transmission Symptom Diagnostics.<br>With the DRBIII®, read Transmission DTC's. Record all DTC's and 1 Trip Failures.<br>**NOTE: Diagnose 1 Trip Failures as a fully matured DTC.**<br>Using the wiring diagram/schematic as a guide, inspect the wiring and connectors. Repair as necessary.<br>Perform the Shift Lever Position Test. If the test does not pass, refer to Symptom test for P0706 Check Shifter Signal.<br>For Gear Ratio DTC's, check and record all CVI's.<br>Most DTC's set on start up but some must be set by driving the vehicle such that all diagnostic monitors have run.<br>**NOTE: Verify flash level of Transmission Control Module. Some problems are corrected by software upgrades to the Transmission Control Module.**<br>**NOTE: Check for applicable TSB's related to the symptom.**<br>Perform this procedure prior to Symptom diagnosis.<br>Continue<br>Go To 2 | All |
| 2 | With the DRBIII®, read Transmission DTC's<br>If any of these DTC's are present, perform their respective tests first.<br>Are the DTC's P0944, P0715, P0720, or P1794 present also?<br>Yes → Refer to the Transmission category and perform the appropriate symptom. If any of these DTC's are present, they will cause a gear ratio error. Perform the test for Loss of Prime first if it is present. Perform 42RLE TRANSMISSION VERIFICATION TEST - VER 1.<br>No → Go To 3 | All |